EXECUTIVE CASE STUDY

QA OPERATIONS & AI GOVERNANCE

# Case Study 1: Enterprise AI QA Governance & BDD Automation

Eliminating the STLC Planning Bottleneck & Accelerating Cross-Border Engineering Delivery

**Melyrose Conciso**
Executive Quality & Operations Leader

**Strategic Impact**
Cross-Border STLC Transformation

STLC CONTEXT

# Context & Operational Bottleneck

**Background:** Managing enterprise software quality at Infor required delivering detailed, execution-ready test cases to a dedicated test automation engineering team in Sri Lanka.

## ⚠ The Core Friction Point

- **STLC Bottleneck:** Manual test case creation choked sprint planning efficiency.
- **Scripting Demands:** Offshore engineers required strict parameters, edge cases, and explicit logic.
- **Resource Drain:** Aligning Product Managers and offshore teams consumed up to **70% of sprint bandwidth**.

## ◎ The Strategic Objective

Build a scalable, AI-directed governance engine that converts raw PRDs into structured **Gherkin features** and **JSON schemas** instantly.

**Key Outcome: Eliminates team idle time without sacrificing enterprise quality standards.**

ARCHITECTURE

# The 4-Tier AI Governance Architecture

PHASE 0

## Shift-Left PRD Audit

AI parses raw requirements to flag specification gaps, edge cases, and GCC compliance risks pre-development.

PHASE 1

## BDD Gherkin Generation

Standardizes business rules into execution-ready `.feature` files with requirement tags ( `@REQ-ID` ).

PHASE 2

## Decoupled JSON Schema

Outputs parameter-driven `.json` files feeding Selenium/Playwright frameworks with minimal maintenance.

PHASE 3

## Human-in-the-Loop

QA Lead audits AI outputs against ISO 25010 quality standards prior to automated pipeline execution.

**Continuous Quality & Compliance Loop**

Ensures 100% traceability from raw product specifications to cross-border automated execution.

ISO 25010 Standard

PROOF OF WORK

# Production-Ready Test Artifacts

Decoupled Execution Model

**remittance_fx.feature** — BDD GHERKIN

```gherkin
@Module:Remittance @REQ-REMIT-02
Feature: FX Rate Lock & Expiration

  @FX @CriticalPath
  Scenario: Lock FX rate for 15 minutes
    Given a bulk payout batch in foreign currency is approved
    When the FX rate lock timer initiates
    Then conversion rate must remain fixed for 15 minutes
```

**test_data_fx.json** — JSON SCHEMA

```json
{
  "tc_id": "TC_FX_002",
  "batch_id": "BATCH_2026_002",
  "locked_fx_rate": 3.67,
  "lock_duration_minutes": 15,
  "expected_status": "FX_RATE_EXPIRED"
}
```

Public Framework Version: Core prompt structure displayed. Enterprise schema adapters available upon request.

PROMPT ENGINEERING

# Master Prompt Architecture (Sanitized)

master_qa_prompt_v2.1.spec

ENTERPRISE AI GOVERNANCE TEMPLATE

```
[ROLE & OBJECTIVE]
You are an Enterprise QA Governance Architect. Parse the provided PRD for feature [MODULE_NAME].

[RULES & GUIDELINES]
1. Identify edge cases, security considerations, and regional compliance rules.
2. Output Gherkin feature scenarios using declarative (business-level) language.
3. Include requirement metadata tags (@REQ-ID, @RiskLevel) for scenario traceability.
4. Generate a separate, matching JSON schema containing input payloads and expected assertions.

[OUTPUT FORMAT]
- Section 1: Requirement Gap Analysis
- Section 2: Gherkin Feature File (.feature)
- Section 3: Parameterized Test Data Schema (.json)
```

BUSINESS IMPACT

# Measured Operational Outcomes

**90%**

STLC TIME REDUCTION

### Sprint Acceleration

STLC test suite design cut from 40 hours to 4 hours per sprint.

**Zero**

DEFECT LEAKAGE

### Shift-Left Quality

Pre-development PRD audits catch requirement gaps before code execution.

**100%**

TEAM UTILIZATION

### Offshore Efficiency

Offshore automation teams receive execution-ready data on Day 1 of sprint.

**3×**

DELIVERY VELOCITY

### CI/CD Automation

Reallocated bandwidth drives rapid CI/CD automation & feature deployment.

LEADERSHIP PLAYBOOK

# How I Apply This to Enterprise QA Organizations

### Embed AI at the Product Gate

Integrate AI PRD parsing during sprint planning so Product Managers and QA finalize specs concurrently.

### Standardize Data Decoupling

Enforce Gherkin + JSON structures across cross-border teams to streamline script handoffs and maintenance.

### Reallocate Team Bandwidth

Shift QA talent from manual typing into framework automation, security governance, and CI/CD pipelines.

**Ready to modernize your QA operations?**
Let's connect to discuss enterprise quality governance.

melyrose.conciso@gmail.com • Dubai, UAE